EG556L, EG556N: AI, Machine Learning and Data Science for the Petroleum Industry

Assessment 2 – Machine Learning & AI
Group coursework

Lateef Akanji (PhD, DIC)

1495 UNIVERSITY OF ABERDEEN

Machine Learning and AI (60 marks)

- Download the ‘Datase01.xlsx’ , ‘best_model.h5’, and ‘random_data_generator.py’ files. The files contain the datasets for the tasks.
- You may use any of the codes provided in the class for your work, but you can also develop your own code, functionalities, or methods. These files must be attached with your submission.
- Download Anaconda and use Jupyter Notebook or Spyder as a template app for this task.
- Endeavour to follow the tips, suggestions and overall format of the template as discussed in the class.
- Submission files – a single zipped file with your group name (e.g., Group_Name) file consisting of:
  - Report of data analysis – page limits are specified where necessary including cover page, graphs and list of references – use arial fonts, 1.5 spacing
  - Jupyter Notebook file (*. ipynb) or Python project file (*.py)
  - Any additional input files such as ‘.csv’.
- Group task execution
  - Each group must agree on members who will execute each of the set tasks.
  - The name of the member(s) of the group who executed each task should be indicated at the top of each page.
  - All members of the group must agree to the content of the submitted zipped files.
  - Hence, a signature page of all members indicating an agreement should be included after the cover page.
  - Only one submission should be made per group.
  - Overall grade marks will consist of 50/50 split between individual contribution and group efforts.

Regression-SVM and random forest – 10 marks

Regression is a technique used to establish the relationship between a dependent variable (y) and one or more independent variables (x). In machine learning, it serves as a supervised learning algorithm for predictive modelling.

1. Using non-linear regression as a supervised learning technique:
   a) Prepare a Python code to analyse the:
      i. Historical production rates (barrels per day, bpd)
      ii. Pressure (psi) – average downhole, average DP tubing, average annulus pressure, average well head pressure

   b) Use the results of your analysis to predict future performance for production rates and pressure by implementing linear regression in your code. Your prediction should cover the period between 2014 and 2024.

2. Use support vector machine (SVM) and random forest to carry out the regression and classification of the historical production rate and pressure. Compare the two outputs and indicate any observable class in the dataset.

3. Discuss how your supervised learning tool can help optimise and improve production strategy for this field.

4. Required dataset for the task – 'Dataset01.xlsx - production'

5. Submission requirement – Python codes – non-linear regression SVM and random forest, input datafiles, report of analysis not more than 5 pages including all figures.

K-means clustering for well log – 10 marks

k-means algorithm can be used to divide data into K clusters, where each data point is assigned to the nearest cluster centroid. The centroids are updated iteratively to minimise the within-cluster variance. Unlike supervised learning where data is labelled, k-means takes unlabelled data and attempts to fragment that into a specified number.

1. Using k-means as an unsupervised learning algorithm coded in Python generate cluster of all the well log measurements to find:
   i. The number of facies
   ii. The different formations in the borehole
   iii. The degree of geological complexities

2. Explain which of the variable properties give an indication of the complexity of the formations present in the borehole using both Elbow Method or Silhouette Score methods.

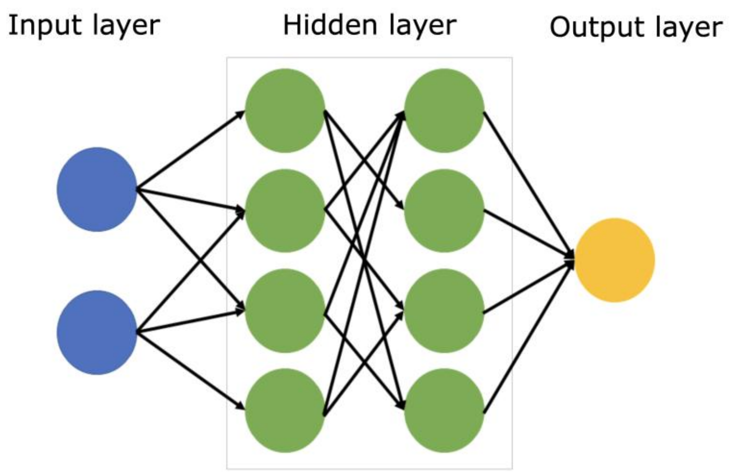
3. Clearly indicate your training data, and test point within your Python code.

4. Discuss the limitations of the use of k-means in relations to spherical (equally sized and shaped) clusters and the presence of any outliers in this datasets

5. Attach your k-means code, your input datafile and report file of not more than 5 pages

Artificial neural network – 10 marks

1. ANN is a supervised learning technique that can model more complex datasets. They handle classification problems with non-linear boundaries Using tensorflow functionalities in Python, generate an artificial neural network model with architecture shown below.



2. The xtrain and ytrain data should be randomly generated using the ‘random_data_generator.py’ code provided. Make sure that the generated data does not change during your analysis.

- Define a simple neural network with one activation function and two inputs.
- For the activation function, evaluate RelU and Softmax with Adam as optimiser.
- Monitor validation loss and find the minimum, stop after 40 iterations without improvement.
- Use the ‘best_model.h5’ file provided.
- Plot the feature space of the Neural Network.
- Use horizontal stack vectors to create x1, x2 input for the model.
- Submit the generated images, the codes and input datafiles.

STOIIP Monte-Carlo calculator – 15 marks

One of the techniques of estimating stock tank oil initially in place (STOIIP) is via the use of Monte-Carlo probabilistic approach.

You are provided with a spreadsheet named 'Dataset' which contains a worksheet 'Montecarlo_STOIIP'. Using the sheet, P10, P50 and P90 STOIIP (N) can be calculated based on Montecarlo simulation with normal distribution-based uncertainties. Specified mean and standard deviations have been used to generate the random values of each variable in the worksheet.

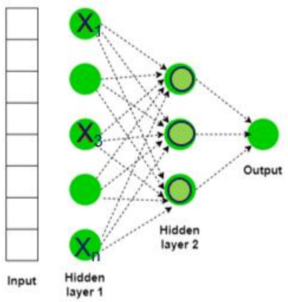
$$N = \frac{7758 A \emptyset h \gamma S_{oi}}{B_{oi}} \ (STB)$$

A=reservoir area [acre-ft]; h=reservoir gross thickness [ft]; Boi=initial oil formation volume factor [bbl/stb]; ϕ=porosity [fraction]; γ =net-to-gross [fraction]; $S_{oi} =$ initial oil saturation

1. Generate:
   i. A Python-based STOIIP Monte-Carlo calculator to automate the process and print the generated outputs and plots.
   ii. Histograms of input variable distributions for area, thickness, initial oil saturation, porosity, net-to-gross and initial oil formation volume factor
   iii. STOIIP Monte Carlo and P10/P50/P90
   iv. STOIIP normal distribution with P10/P50/P90 lines
   v. Tornado charts using normalised weights
   vi. Scatter plots of input variables versus STOIIP
   vii. Include a graphical user interface (GUI) in your calculator
2. Your submission should consist of the Python calculator, the printed output files appended to your report and any spreadsheet used.

AI model for predicting oil recovery – 15 marks

Predicting the performance of an enhanced oil recovery (EOR) campaign is a critical step in a mature oilfield operation. AI model can be used to obtain insight into the performance of a producing asset by providing the most appropriate option out of miscible gas, chemical, thermal or microbial EOR methods. Using the data provided from world successful EOR projects, develop a three-layer feed-forward neural-network and fuzzy-logic based AI screening algorithm in Python with architecture similar to the figure below. Parameters to be used for the screening process include API gravity, oil saturation, depth, permeability, porosity, and viscosity.



1. Use skfuzzy functionality – fuzzy logic toolbox – in Python.
2. Prepare the simulated training, prediction, and error data using random uniform method by creating an array with random samples from a uniform probability distribution of given low and high values corresponding to API gravity, oil saturation, depth, permeability, porosity, and viscosity data. This analysis should be done for the training and prediction process.
3. Generate the learning curves and training and prediction graphs for all the parameters.
4. Use testing data from the published work of Ramos and Akanji, 2017 available here: EOR-screening, for the test-prediction analysis. This analysis should be done for the testing and prediction process and report associated error – such as root-mean-square error (RMSE), non-dimensional error index etc. Include a graphical user interface (GUI) in your EOR screening tool.
5. By applying linguistic rules on your fuzzy logic model and combining it with the learning capability of Neural Network, suggest an appropriate EOR technique for the predictive performance analysis.
6. Submit the generated graphical outputs, the codes and input datafiles used in your work.